Designing for Abductive Meaning-Making: An Explainable Interface for GenAI-Mediated Ill-Defined Problem Solving

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The growing integration of GenAI in education creates a tension between efficient output production and learners' ability to understand the reasoning behind AI-generated responses, particularly in ill-defined problem contexts. This study reframes explainability not as an intrinsic property of AI systems, but as an interactional process emerging among learners, AI outputs, and interface structures. Drawing on Peircean semiotics, it conceptualizes abductive reasoning as a key mechanism through which learners interpret and reconstruct the meaning of GenAI outputs under conditions of epistemic opacity. Adopting a Research through Design approach, the study develops an Explainable Interface prototype, the Abduction Journal, designed to externalize abductive reasoning through proposition-based interaction and relational structuring. Three participatory design workshops involving 11 participants informed three design principles: acknowledging interpretive instability, expanding the space of interrogation, and grounding AI outputs in problem context. These principles were translated into an interactive canvas and evaluated through user testing with five participants. The findings suggest that predefined propositions can lower the threshold for entering reasoning activity, while relational judgments between propositions remain difficult for users to construct. A tentative pattern was observed between the coherence of participants' reasoning structures and the quality of their subsequent problem reframing. The study contributes an interface-oriented account of explainability grounded in abductive meaning-making and proposes abductive reasoning as an additional competency dimension for Computational Thinking in GenAI-mediated ill-defined problem contexts.

CCS CONCEPTS • Human-centered computing • Interaction design • Artificial intelligence

Additional Keywords and Phrases: Explainable AI, Abductive Reasoning, Human-AI Interaction, Meaning-making, Semiotic Interaction, Ill-defined Problem Solving, RtD

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1 INTRODUCTION

1.1 Problem Statement

In recent years, interest in the application of artificial intelligence in education has grown rapidly [11]. Although GenAI tools provide learners with unprecedented support and efficiency, they also introduce structural tensions into learning processes. One of the most prominent concerns is epistemic opacity: GenAI systems operate through probabilistic and largely non-transparent mechanisms, making their internal reasoning processes difficult to inspect. As a result, students may encounter outputs that appear coherent and functionally complete, while lacking access to the reasoning pathways through which these outputs were produced. This development raises fundamental questions about the nature of understanding in computing education.

These tensions become particularly salient in contexts involving uncertainty and ill-defined problems. Unlike well-defined tasks with clear inputs and solution paths, ill-defined problems require learners to engage in exploratory reasoning, hypothesis generation, and iterative problem framing. However, when GenAI systems directly provide seemingly complete solutions, learners may bypass these formative reasoning processes. Under such conditions, AI opacity risks weakening a core dimension of computing education: the abductive generation, evaluation, and revision of hypotheses.

Current discussions on AI in education have largely focused on issues such as academic integrity, tool integration, AI literacy, and assessment reform [2, 18, 35, 38]. While these concerns are important, they often remain at a technical or policy level, paying comparatively limited attention to the epistemic structures that underlie computing practice. Computational thinking is commonly framed in terms of abstraction, decomposition, and algorithmic design, yet such definitions tend to underemphasize the interpretive and hypothesis-driven dimensions of programming, particularly in environments characterized

by uncertainty and incomplete information. The conceptual role of abductive reasoning in AI-mediated computing education remains underexplored.

1.2 Research Question

In this study, abductive journaling refers to an interface-supported practice in which learners record, compare, and revise hypotheses about AI-generated outputs during problem solving. Building on the theoretical tensions identified above, this study formulates the following primary research question:

How can an Explainable Interface centered on abductive journaling mediate the explainability of GenAI and support learners' meaning-making in ill-defined problems?

To systematically address this question, three interrelated sub-questions are established at progressively concrete levels of the research process:

SQ1: What theoretical and design principles should inform the construction of an Abductive Journaling interface for GenAI-mediated ill-defined problem solving?

SQ2: How can abductive reasoning processes be externalized and structured through interface interaction in GenAI-mediated learning contexts?

SQ3: In what ways does interaction with the Abduction Journal interface influence learners' engagement with GenAI outputs in ill-defined problem contexts?

1.3 Scope and Delimitations

This study does not aim to explain or modify the internal mechanisms of GenAI models. Instead, it focuses on interface-level design and qualitative analysis of meaning-making processes, examining how interaction structures can support learners' engagement with AI-mediated ill-defined problems under conditions of epistemic opacity.

2 BACKGROUND

2.1 Transparency Problem in AIEd

With the rapid emergence of accessible GenAI technologies, students at all levels of education are increasingly able to use these tools throughout the learning process to support the completion of academic tasks. However, across educational systems worldwide, a new learning crisis is gradually becoming visible: with the support of AI, students can efficiently produce outcomes that meet formal requirements, yet often struggle to trace their own thinking processes or to explain the reasoning that underpins these results. As one U.S. high school student bluntly remarked in The Atlantic: "When you can hand your thinking over to a machine and still get an A, who cares whether you actually learned anything?" [28]

This tension is closely related to what has been widely discussed as the Transparency Problem in AI systems. GenAI systems typically operate through probabilistic and opaque mechanisms, producing outputs whose internal reasoning processes are difficult or impossible for users to directly inspect. As a result, learners may encounter responses that appear coherent, fluent, and functionally complete, while lacking access to the inferential pathways through which these outputs were generated. This epistemic gap challenges traditional assumptions about how understanding is formed and evaluated in computing education.

The issue becomes particularly salient in computing education contexts, where learning tasks often involve abstract reasoning, judgment under uncertainty, and the iterative reframing of problems [14]. In such contexts, students are not merely expected to arrive at correct answers, but to develop an understanding of why particular solutions work and under what conditions they remain valid. When AI systems intervene by providing ready-made outputs, these reasoning processes risk being obscured rather than supported.

In principle, all stages of AI decision-making should be open to scrutiny, interpretation, and explanation [8]. In practice, however, the opacity of many AI systems does not result from intentional concealment, but from intrinsic technical constraints—particularly in models based on deep neural networks. Under conditions of such deep opacity, requiring AI systems to provide fully transparent, causally faithful explanations is often unrealistic.

Recent work by Daniels and colleagues therefore argues that the central challenge is not whether AI systems themselves can be made fully transparent, but whether learners possess the competencies required to engage meaningfully with opaque systems, specifically, the capacity for interrogability [14]. Interrogability refers to an operational form of epistemic engagement, including the ability to question AI outputs, probe boundary conditions, and evaluate the credibility and scope of AI-generated responses. From this perspective, the transparency problem is reframed as an educational challenge rather than a purely technical one.

2.2 Limitations of Existing XAI Approaches

Reframing the issue of transparency as an educational challenge does not imply that technical considerations can be set aside. The explanatory resources accessible to learners in their actual use of GenAI remain, to a large extent, shaped by the technical capacities of XAI systems. In recent years, the field of XAI has proposed a range of methods aimed at enhancing the interpretability and auditability of AI systems, including LIME [27], LRP [1] and DeepLIFT [32]. Nevertheless, the practical effectiveness of XAI in supporting human understanding remains limited. Empirical studies indicate that even for relatively simple tasks, many users struggle to meaningfully interpret and apply the explanations provided by XAI systems [17]. Chromik and Butz [3] further argue that technical explanations often fail to translate into genuine understanding at the user level, particularly when their presentation does not account for users' goals, prior knowledge, or interpretive practices.

Nguyen and Zhu [20] highlight that a central limitation of many XAI systems lies in how user needs are conceptualized: explanatory requirements are often inferred from a technical perspective rather than grounded in empirical investigations of how users actually make sense of AI outputs. As a result, XAI explanations may be technically accurate while remaining misaligned with the epistemic demands of real-world decision-making and learning contexts.

Within computational education, these limitations are further amplified, and the dominant competence frameworks in the field themselves presuppose a particular characterization of the problem space. Since its introduction by Wing, *Computational Thinking* (CT) has been grounded in core competencies such as abstraction, decomposition, and algorithmic design [9, 41]. This framework assumes, to some extent, a problem space that can be structured, where tasks are understood to involve identifiable initial and goal states. However, the increasing integration of GenAI into computational education introduces a qualitatively different form of learning challenge: AI outputs are generated through probabilistic prediction rather than traceable causal logic, rendering their meaning fundamentally opaque to learners and not directly accessible through conventional system logic or formal reasoning [14]. In such contexts, the learner's cognitive task shifts from executing predefined algorithms to reconstructing the meaning of AI outputs interpretively in the absence of visible reasoning pathways.

2.3 Explainable Interfaces as a Design-Oriented Response

Against this backdrop, recent work has shifted attention from algorithm-centered explanation methods toward the design of *Explainable Interfaces* (EI), positioning the interface as an intermediary layer between AI systems and human understanding [21]. Unlike XAI approaches that attempt to reconstruct or expose the internal mechanisms of models, EI is less concerned with what to explain than with how explanations are conveyed—specifically, how interface structures, interaction design, and modes of presentation shape the effectiveness of explanations in situated user contexts.

From a human-centered design perspective, the central premise of EI is that explanation is not a static output produced by an AI system, but a process that unfolds progressively through interaction. This premise takes diverse forms in design practice. For instance, the matrix-based interface proposed by Chromik et al. [4] enables users to sort, filter, and modify the feature data of 16 loan applicants, supporting the gradual construction of understanding through comparative and counterfactual reasoning. The chess interface developed by Das and Chernova [6] adopts an organic information architecture in which explanations emerge dynamically through interaction rather than being predefined. Similarly, QuestionComb by Sevastjanova et al. [31] frames the annotation of linguistic phenomena as a gamified activity, allowing users to develop an understanding of a language model's classification logic through incremental labeling. Guo et al. [10] extend this approach by enabling users not only to receive explanations but also to contribute their own decision rules, thereby turning explanation generation into a product of human–AI collaboration. Further, Villareale et al. [39] demonstrate, through their exploration of simulation games as a form of EI, that embedding explanations within action–feedback loops can support users in progressively constructing an understanding of AI system behavior—an understanding achieved through participation rather than passive reception. Across these examples, a consistent design logic emerges: the explanatory quality of an interface depends not only on the completeness of the information it provides, but also on whether it creates conditions that meaningfully support users' active engagement and judgment.

However, conceptualizing explanation as an interactional property addresses only the structural features that interfaces should provide; it does not account for the nature of the cognitive activity users engage in when they operate, evaluate, and revise within these conditions. What kind of cognitive process is taking place in such interactions, and what internal logic characterizes it? Existing EI frameworks primarily focus on how interfaces organize the presentation and interaction of explanations, rather than on the generative mechanisms underlying the act of explaining itself. This suggests the need for a theoretical perspective capable of describing the internal structure of meaning-making processes.

2.4 Research Gap

A review of existing research suggests that both XAI and Explainable Interfaces have advanced AI interpretability in distinct ways. However, to some extent, they share an underlying Cartesian dualist epistemology: explanatory activity is framed as occurring between a knowing subject (the learner) and an object system (GenAI) that produces outputs, with meaning located on the side of the subject and AI outputs treated as objects awaiting interpretation. While this assumption may be operationally useful at a technical level, it constrains a fuller account of the nature of explanation—particularly in ill-defined problem contexts. In such settings, the learner's central challenge is not simply to obtain sufficient explanatory information, but to actively construct, externalize, and revise interpretive hypotheses in the absence of visible reasoning pathways. Existing frameworks offer limited means to characterize the internal structure of this activity and, consequently, provide little theoretical grounding for interface designs intended to support it.

This study addresses this gap by drawing on Peircean semiotics and its theory of abductive reasoning as a conceptual framework. The subsequent section on the theoretical framework elaborates how the triadic structure of signs provides an epistemological basis for moving beyond the dualist assumption outlined above, and how this foundation may be translated into concrete principles for interface design.

3 THEORETICAL FRAMEWORK

3.1 Why Peircean Semiotics: Ontology and Epistemology

> "*This tradition of Poinsot-Locke-Peirce, unlike that of Saussure, does not take its principal and almost exclusive inspiration from human language and speech. It sees in semiosis a broader and much more fundamental process, involving the physical universe itself in human semiosis, and making of semiosis in our species a part of semiosis in nature*" -*John Deely*[12]

Deely's distinction between two semiotic traditions points to fundamentally different ontological positions. Traditions in philosophy of science and epistemology have typically taken as their point of departure the question of "how knowledge is possible," a concern that also underlies the philosophical work of Charles Sanders Peirce. This question establishes both the starting point and the boundary for human efforts to understand and interpret AI. Peirce's reformulation of Kant's transcendental logic through semiotics indicates that his epistemology is, at its core, inseparable from his theory of signs.

In the Saussurean tradition, a sign consists of the *signifier* and the *signified*, with meaning arising from differential relations among signs within a linguistic system. This constitutes a Cartesian dualist structure: the meaning of a sign depends on the presence and absence of other signs within the same system, while its relation to the physical world remains, in principle, external. In contrast, Peirce holds that new knowledge is logically determined by prior knowledge, with each element of new understanding derived from inferences based on previous cognition [24]. If knowledge is understood as fundamentally inferential, involving logical relations among different cognitions, then it may be conceived as a structure constituted by interconnected signs. In this sense, human cognition is inherently dependent on signs and cannot proceed independently of them; thinking itself does not extend beyond the domain constituted by sign processes. For Peirce, thought is essentially public: it is not confined within individual minds but exists within the shared symbolic structures used for communication. Thought thus possesses an external dimension; only when individual minds function as carriers of signs does this externalized thought gradually become internalized [40].

Peirce's theory of signs is triadic at multiple levels. Most fundamentally, he conceives the sign as comprising the *Representamen*, the *Object*, and the *Interpretant*. This triadicity carries significant ontological implications: meaning does not pre-exist in the representamen itself, nor is it embedded within the object it refers to; rather, it emerges through the dynamic relations among these three elements. The interpretant, in particular, initiates an ongoing process of interpretation that is inherently generative and transformative, keeping the triadic structure open and fluid. This distinguishes it from simple stimulus–response models and situates meaning within an unfolding process of interpretation. Each interpretant can in turn function as a new sign, enabling an infinite recursive chain of interpretation that is central to the concept of semiosis.

Peirce's account of signs and meaning provides a basis for moving beyond the Cartesian framework that informed early AI design [7]. Thinking can be understood as a dynamic process that relates the three elements of semiosis. The epistemological implications of this ontological stance are particularly relevant for analyzing GenAI-mediated learning contexts. The Saussurean dualist assumption—where an interpreting subject confronts a sign object that carries meaning—reveals its limitations in such settings. It locates meaning on the side of the subject and treats GenAI outputs as objects awaiting

interpretation, thereby presupposing a stable subject–object boundary. In practice, however, no such stable boundary exists among the learner, GenAI outputs, and the interface structures through which they interact. Meaning is not extracted from AI outputs; rather, it is progressively generated through ongoing interaction between the learner and those outputs. The Peircean triadic framework offers an alternative ontological starting point: meaning emerges through relational processes rather than residing in any single component. This allows the learner, GenAI outputs, and interface conditions to be understood as participating within a unified semiotic process, without presupposing a fixed subject–object division.

3.2 Semiotic Status of GenAI Outputs

On the metaphysical foundation of his semiotic theory, Peirce proposed three phenomenological categories to describe modes of being in experience: *Firstness* (the category of quality), *Secondness* (the category of fact), and *Thirdness* (the category of relation) [24].

As shown in Table 1, from this perspective, Secondness constitutes the irreducible grounding of experience. It is present in every contingent event, manifesting as relations of action and resistance that cannot be suspended or bypassed by the subject. However, reducing cognition to this level alone is insufficient. Cognition, as a process mediated by sign activity, cannot be equated with the direct reaction of one object to another in the world—an account that corresponds only to Secondness. A more adequate characterization situates cognition within Thirdness: an interpretive process unfolding through semiosis, in which something is understood as something within a structured relation.

Within this framework, this study conceptualizes GenAI outputs as exhibiting a form of *quasi-semioticity*. In this sense, such outputs may be described as artifacts that formally trigger interpretive activity while lacking the phenomenological conditions required for a fully realized sign process. Large language models may simulate aspects of Thirdness, insofar as neural networks extract patterns from training data; however, they fundamentally lack the anchoring of Secondness. Their outputs do not refer to external objects through causal or experiential connections, nor are they subject to resistance or counteraction from the physical world. Consequently, they are unable, in semiotic terms, to distinguish between hallucination and truth.

This distinction can be illustrated through the example of a red traffic light. Humans encounter all three categories simultaneously: redness as a quality (Firstness), the physical constraint imposed by the signal (Secondness), and the interpretation of "stop" within a socio-cultural system (Thirdness). By contrast, although large language models have processed the sign "red light" countless times in textual data, they have never encountered either the qualitative possibility of redness or the resistance of an actual traffic situation.

This quasi-semiotic condition constitutes a deeper epistemological source of the opacity associated with GenAI outputs and provides a semiotic account of the structural difficulty learners face in constructing meaning. What learners require, therefore, is not simply more technical explanation, but conditions that support the active generation and testing of interpretive hypotheses. This process of active generation is precisely what the next section will address through the concept of abductive reasoning.

Table 1 Human Experience and GenAI Through Peirce's Categories

Category	Semiotic Meaning	Human Experience Example	GenAI Output
Firstness	Quality, possibility, and immediate experience	Perceiving redness	Absent. LLMs do not experience qualities such as redness.
Secondness	Fact, action, and resistance encountered in the world	Encountering a red traffic signal as a fact of the situation that demands a response	Absent. Outputs are not grounded in causal or experiential encounters with reality.
Thirdness	Interpretation through relations, rules, and mediation	Understanding it as "stop" within a socio-cultural system	Partially simulated through statistical pattern extraction from training data.

3.3 Abductive Reasoning

"*Abduction, the process whereby alone new ideas are seized upon—ideas further to be developed deductively and tested inductively, beginning again the cycle—or, rather, spiral—is first of all a phenomenon of nature. It works with constructed signs, but not only with constructed signs, and not with constructed signs first of all.*" *-John Deely* [12]

Within the traditions of philosophy of science, reasoning has long been differentiated into three fundamental forms, a distinction that carries significant implications for understanding processes of meaning-making. Inductive reasoning derives general patterns from observations, while deductive reasoning derives conclusions from established premises. As Peirce observed, "neither deduction nor induction contributes the smallest item of new information to the final conclusion of the inquiry" [23]. Both forms of reasoning thus remain confined to what is already given.

It is precisely in this respect that abductive reasoning assumes epistemological distinctiveness. In his early work, Peirce referred to this process as *hypothesis*, using it to characterize the origin of creative reasoning in scientific inquiry [26]. Abduction begins with a surprising or anomalous observation and, through a more complex inferential movement, introduces explanatory content that is not contained in the premises but remains consistent with them [23].

The canonical formulation of abduction appears in Peirce's 1903 Harvard lectures [25]: a surprising fact C is observed; if A were true, C would be expected; therefore, there is reason to suspect that A is true. When an inquirer encounters an unexpected phenomenon, however subtle, it comes into tension with prior experience, prompting the search for a premise under which the phenomenon would no longer appear surprising, thereby alleviating the cognitive tension it produces.

Abduction neither derives conclusions from given rules nor generalizes from repeated instances; rather, it generates a hypothesis that renders the observed phenomenon intelligible. The historical formulation of gravity illustrates this process: the concept of "gravity" did not emerge directly from observation, but as a hypothesis introduced to explain observed phenomena.

This spiral structure highlights the distinctive nature of abduction: it is the most flexible mode of reasoning, operating outside the strict determinism associated with deduction and induction, and is therefore often regarded as central to critical thinking, cognitive flexibility, and creativity [16]. From the perspective of Peircean semiotics, abduction defines the dynamic process through which interpretants are provisionally constructed within semiosis: the reasoner begins with a given representamen, generates candidate interpretants, and subjects them to further semiotic activity for evaluation and revision. Each interpretant that achieves temporary stability becomes the starting point for subsequent abductive movement.

As discussed earlier, GenAI outputs, as quasi-semiotic artifacts lacking anchoring in Secondness, are generated through the inductive aggregation of statistical patterns rather than through traceable inferential pathways. This implies that learners cannot reliably derive stable causal regularities from such outputs through induction, nor can they validate conclusions through deduction based on accessible premises, as the structural conditions required for both modes of reasoning are absent. Instead, learners must begin with the given output and abductively infer what kind of problem representation or implicit premises would render it intelligible. This abductive movement constitutes the central cognitive mechanism through which learners engage in meaning-making in GenAI-mediated contexts, and it occupies a role that cannot be structurally replaced by either induction or deduction. The situational conditions under which this abductive process unfolds will be further examined in the following section on ill-defined problems.

3.4 Ill-Defined Problem

Ill-defined problems characterize the structure within which learners engage in meaning-making when encountering GenAI outputs, clarifying why this process cannot be reduced to algorithmic execution nor substituted by AI-generated results. According to Allen Newell and Herbert Simon, a problem space consists of (1) an initial state, (2) a final desired state, and (3) operators [19].

Based on the clarity of these elements, problems are commonly distinguished as well-defined or ill-defined. Typical mathematics examination problems exemplify well-defined problems: their initial conditions are explicit, goal states are clearly specified, operators are stable and predictable, and solutions are typically derived through deductive reasoning. In contrast, ill-defined problems exhibit markedly different characteristics: the available information may be complex, incomplete, or ambiguous; the goal state may be under-specified or admit multiple acceptable outcomes; and the pathways linking the initial and goal states are often unknown, non-linear, and highly dependent on contextual judgment [33]. Under such conditions, the central challenge lies not in executing predefined algorithms, but in constructing and revising problem representations under uncertainty. Even in well-defined problems, problem solving involves the construction of internal representations that connect the initial state, goal state, and available operators. However, these representations are typically stable and supported by established procedures, making deductive reasoning the dominant mode of inquiry.

Ill-defined problems introduce further complexity. Consider the question: "A hunter shoots at a tree. There were three birds sitting on the branches. How many birds are left?" While it appears to be a quantitative problem, the word "left" carries dual meanings—both "remaining" and "having departed." Whether the hunter successfully hits the target is unspecified, and the birds' reactions to the gunshot may vary depending on the interpreter's experience and cultural background. These divergent

interpretations are not trivial complications; rather, they reveal a fundamental feature of ill-defined problems: different interpreters may extract different initial states from the same problem statement and anticipate different goal states.

Many activities in computational education, such as software design and requirements analysis, are ill-defined because they involve incomplete information, multiple acceptable outcomes, and context-dependent judgments.With the increasing integration of GenAI into problem-solving contexts, the structural challenges of ill-defined problems are further intensified. GenAI systems can function as executors of operators, rapidly generating candidate solutions. However, due to the quasi-semiotic nature of their outputs, the learner's task extends beyond "how to solve" toward "how to understand where this solution comes from." Large language models can simulate textual patterns associated with induction and deduction, yet this simulation constitutes a reduction of the triadic dynamics of semiosis: their outputs do not structurally encode genuine inferential pathways—not because such pathways are concealed, but because they are not generated in the first place.

In AI-mediated ill-defined problems, learners face two forms of uncertainty: ambiguity within the problem itself and opacity in AI-generated outputs. Abductive reasoning becomes the mechanism through which learners reconstruct possible interpretations of both. Explanation and problem solving therefore become intertwined aspects of the same meaning-making process.

3.5 Explanation and Understanding

Given that semiosis is indefinitely extensible, the question arises as to how understanding achieves a "good enough" point of stability in practice. From a semiotic perspective, explanation is not a terminal operation but an ongoing process of sign activity. As emphasized by John Deely, the interpretant is both the outcome of one semiotic process and the starting point of another [12]. Once articulated, an interpretant itself becomes a sign capable of generating further interpretants. Explanation thus unfolds as a spiral chain of semiosis, in which meaning is continuously produced rather than statically contained. This structure also foregrounds the dialogical nature of explanation: effective explanation is not a static output transmitted unidirectionally from an AI system to a learner, but an interactive process that develops through iterative questioning and response [14, 15].

At the same time, the potential infinitude of explanation does not imply that understanding can expand without limit. Effective explanation must achieve a form of provisional stability within a given context, and the conditions under which such stability emerges require theoretical clarification. Van Fraassen's pragmatic theory of explanation provides a useful analytical lens in this regard [37]. Within this framework, explanation is understood as an answer to a "why" question, and any such question comprises three elements: the phenomenon to be explained, a contrast class, and a relevance relation. The contrast class is defined by a particular community of inquiry and delimits the scope of what the explanation is expected to address; the task of explanation is not to exhaust all possible causes, but to clarify "why this rather than that." What counts as a satisfactory explanation therefore depends on the inquirer's interests, background knowledge, and investigative goals, rather than on any intrinsic property of the phenomenon itself [5]. The adequacy of an explanation is not determined by complete transparency, but by whether it satisfies the relevance relations specified by the contrast class.

On this basis, a conceptual model can be constructed to describe GenAI-mediated problem-solving processes (Figure 1). The model delineates two semiotic trajectories that develop in parallel and eventually converge. The first trajectory concerns the problem statement: starting from an initial problem representation, the learner generates interpretants through personal knowledge and abductive hypotheses. Each interpretant reconstructs the understanding of the problem at a new level, forming an upward progression of meaning-making. The second trajectory concerns the GenAI output: as a quasi-semiotic intermediary sign, the AI-generated output enters the interpretive process, prompting the learner to generate abductive hypotheses that reconstruct the implicit premises under which the output becomes intelligible within a given contrast class. These two trajectories mutually inform and constrain one another through ongoing interaction: the learner's evolving understanding of the problem shapes the interpretation of the AI output, while abductive reconstruction of the AI output, in turn, contributes to further clarification of the problem representation.

The two trajectories ultimately converge at a *Higher-Order Object*, forming a temporarily stabilized structure of understanding that is sufficient for the current context. This structure does not constitute a final interpretant, but rather a provisional point of stability defined by contextual adequacy—jointly determined by the contrast class and relevance relations established by the learner. In this sense, explanation and understanding may be conceptualized as processual outcomes that gradually stabilize under conditions of uncertainty, rather than as predetermined endpoints. The systematic questioning capacities required of learners in this process will be further addressed at the level of design principles.

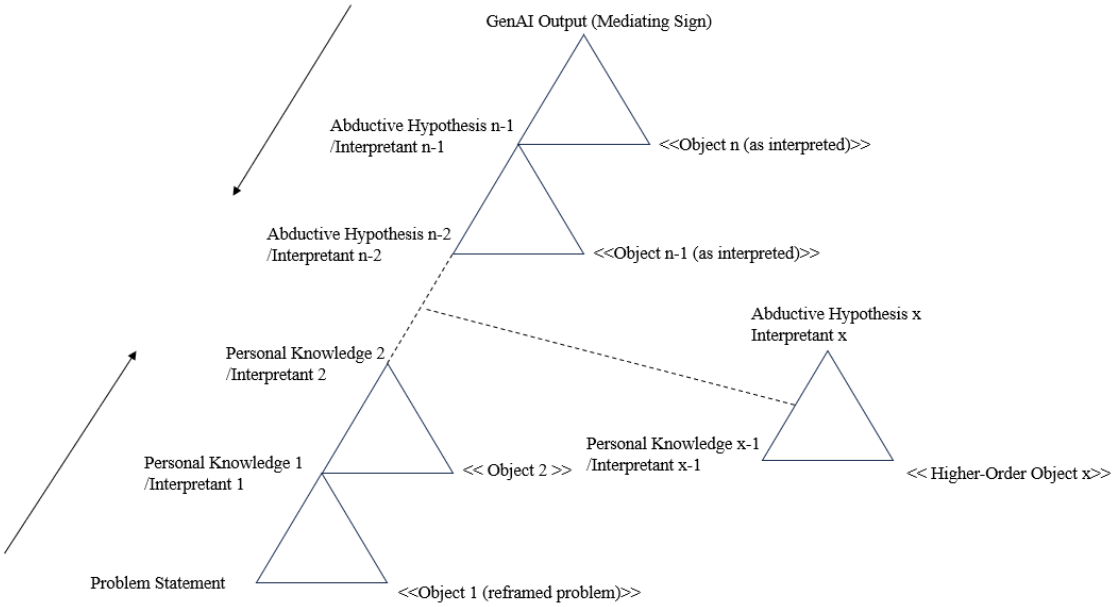


Figure 1. A semiotic model of abductive integration in GenAI-mediated problem solving.

4 METHOD

4.1 Research through Design

This study adopts Research through Design (RtD) to organize the research process, drawing on the framework proposed by Zimmerman et al. as a methodological reference [42]. In this study, RtD is used to structure how the research progresses from a theoretical framework through empirical inputs, design translation, and reflective evaluation, ultimately generating analyzable research material.

Within the RtD orientation, design is understood as an exploratory practice through which researchers engage with problem spaces in situated contexts. Through the conception, construction, and iteration of artefacts, initial assumptions can be explored and revised. Knowledge is not treated as a pre-existing object to be verified, but as something that emerges through cycles of design action and reflective analysis.

Guided by this approach, the study conducts design interventions in learning contexts mediated by GenAI, focusing on the construction and evaluation of an explanatory interface prototype referred to as the Abduction Journal. The design process is not conceived as a linear progression toward a predetermined outcome, but as a series of design actions through which theoretical constructs are progressively translated into interaction structures, and through which knowledge emerges from the sustained dialogue between framework and practice. In this sense, RtD primarily functions as a mechanism for organizing the research process and coordinating methods.

As the overarching methodological framework, RtD provides the structure through which theoretical concepts are translated into design interventions and subsequently evaluated, thereby supporting the investigation of all three sub-questions (SQ1–SQ3).

4.2 Participatory Design Workshop

This study employs Participatory Design Workshops (PD Workshops) as the primary method for user data collection. PD seeks to involve diverse stakeholders in the design process, positioning them not merely as research subjects but as co-creators of design outcomes [34]. As a human-centered approach, PD extends user involvement beyond the evaluation of design solutions to include active participation in the generation of ideas [30].

The theoretical framework of this study engages with abstract concepts such as *semiosis* and *abductive reasoning*. How these concepts can be operationalized at the level of interface design cannot be determined through theoretical deduction alone. Workshops provide a situated context in which theoretical assumptions can be brought into interaction with real use scenarios, enabling the researcher to examine the extent to which these assumptions align with actual user needs.

Moreover, the re-interpretive dynamics inherent in participatory design workshops resonate with the semiotic orientation of this study. Participants' responses to design concepts constitute acts of interpretation, whereby each proposed idea, as it is received and modified by others, generates new interpretants. In this sense, the workshop functions not only as a means of data collection but also as a process through which discrepancies between the researcher's assumptions and participants' perspectives can be identified and refined.

Within this study, the workshop serves as a site for calibrating and concretizing theoretical premises, while producing design directions that have been examined in context. At the level of implementation, although a semiotic perspective and user-centered HCI principles informed the overall structure of the workshop, these theoretical constructs were not explicitly introduced to participants during the sessions. Instead, the semiotic framework and its associated critiques are applied in the subsequent analysis.

The workshop findings primarily contribute to SQ1 by identifying user needs, expectations, and design directions that inform the development of design principles for the Abduction Journal interface.

4.3 Prototyping

Prototyping is a central method in design research, aimed at exploring and refining interaction concepts through experiential engagement. It enables abstract design principles to be materialized into interactive interface structures, thereby allowing user experience to be examined at an early stage. Within the RtD framework, prototypes are not treated as finalized products but as evolving artifacts. Their function is to externalize theoretical concepts into concrete interaction mechanisms, making them observable, usable, and analyzable as research material. As Saffer notes, prototyping does not merely present a design outcome; it serves as a core medium through which designers learn, communicate, and make discoveries during the design process [29].

In this study, prototyping fulfills two interrelated roles. First, it operationalizes the design principles of the *abduction journal* by translating workshop-calibrated insights into specific interface structures and interaction patterns. Second, as an interactive research artifact used in user testing, the prototype is not intended to achieve technical completeness; rather, it is designed to maintain sufficient structural stability to elicit and capture user behaviors that are analytically meaningful.

Throughout the project, the prototype is maintained at a low-fidelity level. This choice helps sustain the research focus on interaction structures and the support of reasoning processes, rather than on visual refinement. At the same time, low-fidelity prototypes tend to encourage participants to provide direct feedback on the functional logic of the *abduction journal*, rather than on aesthetic aspects of the interface. In addition, AI functionality within the prototype is implemented using a Wizard of Oz (WoZ)-like approach, whereby simulated outputs are manually generated by the researcher during testing sessions rather than produced by an actual AI system. This strategy reduces development overhead while enabling rapid and flexible simulation of human–AI interaction.

The prototyping process primarily addresses SQ2, as it translates the design principles derived from theory and workshops into concrete interaction mechanisms intended to externalize abductive reasoning.

4.4 User Test

Throughout the design research process, *user testing* is employed as the primary method for prototype evaluation. In this study, the central aim of user testing is not to assess the visual design or technical completeness of the interface, but to examine whether abductive reasoning appears during actual use of the interface.

Grounded in human-centered design principles, as emphasized by Don Norman [22], observing users performing tasks in situated contexts is an effective way to identify barriers to understanding and structural interaction issues. Accordingly, this study investigates whether the prototype's interaction structure enables users to exhibit characteristics of abductive reasoning within near-authentic usage scenarios. User testing is positioned as a *formative evaluation*, with the goal of collecting qualitative evidence to examine whether the interaction mechanisms give rise to the theoretically anticipated processes of meaning-making, rather than to quantify learning outcomes.

In terms of design, the testing procedure combines situated tasks with post-task interviews. Participants were asked to interact with the prototype interface to interpret and reason about an AI-mediated problem scenario, while engaging in *think-aloud* protocols to externalize their reasoning processes. Immediately following task completion, semi-structured interviews were conducted, focusing on three key aspects: participants' interpretations of AI outputs, changes in understanding before and after using the tool, and their approaches to reformulating queries to the AI system.

It should be noted that, due to certain accessibility limitations of the interface, some participants required operational guidance from the researcher during testing. This intervention was restricted to procedural support rather than conceptual

guidance; however, it nonetheless constitutes a limitation of the study. As a result, the testing process more closely resembles researcher-assisted observational inquiry rather than fully independent user testing.

The user testing phase contributes to both SQ2 and SQ3. It examines whether the interaction structures successfully externalize abductive reasoning processes and explores how engagement with the prototype influences participants' interpretation of GenAI outputs in ill-defined problem contexts.

4.5 Ethical Considerations

The ethical considerations of this study primarily concern the handling of participant data and the externalization of reasoning processes during interaction with generative AI systems. All participants took part voluntarily and provided informed consent prior to participation, having been informed of the study's purpose and scope. Participants in both workshops and user testing sessions remained anonymous, and no personally identifiable information was collected.

All data gathered during the study—including screen recordings, canvas screenshots, think-aloud audio recordings, and interview notes—were anonymized prior to analysis. Data collection, storage, and processing were conducted in compliance with the General Data Protection Regulation (GDPR).

Although the study concerns generative AI, participants did not directly interact with a live AI system during user testing. Instead, AI-mediated interaction was simulated through predefined or researcher-generated outputs. This reduced AI-specific risks related to biased real-time generation, opaque system behavior, and the exposure of participant data to external AI services.

As the study focuses on interface structures and reasoning processes rather than individual performance or learning outcomes, the potential evaluative pressure on participants is minimized. Participants retained the right to withdraw from the study at any time without consequence. Given the exploratory nature of the prototype-based user testing and its low-risk profile, the ethical risks associated with participation are considered minimal.

5 WORKSHOPS

5.1 Data Collection

Three participatory design workshops were conducted at Uppsala University between March and April 2025, involving 11 participants in total. Participants were graduate students from HCI, Computer Science, Information Systems, and related fields, recruited through convenience sampling from the researcher's academic network. Participation was voluntary; all participants provided informed consent before the workshops and remained anonymous throughout the study (see Appendix A.1).

Each workshop lasted approximately 90 minutes and consisted of two functionally connected parts. Part 1 provided participants with situated experience of interacting with GenAI systems, enabling their later design reflections to be grounded in concrete encounters with AI-generated outputs rather than abstract assumptions. Although this phase included think-aloud protocols and group discussion, its data were not included in the formal analysis; instead, it served as a preparatory stage for activating participants' awareness of AI-generated ambiguity and explanation needs.

Part 2 used this shared experiential context as the basis for a participatory design activity focused on the Abduction Journal. Data for the subsequent analysis were generated primarily through an adapted 6-3-5 Brainwriting approach [36]. Participants first wrote initial interface ideas independently and then iteratively responded to, modified, or extended ideas produced by others. This written relay structure supported both independent idea generation and progressive elaboration, producing a traceable corpus of design proposals, annotations, and revisions. Detailed workshop procedures and complete brainwriting records are provided in Appendix A.3.

The adapted Brainwriting procedure is methodologically important because it preserves both the content of participants' design ideas and the process through which those ideas were interpreted by others. This makes the workshop data suitable for analysing recurring design concerns rather than only isolated suggestions. The recorded sequence of initial ideas, annotations, and revisions further supports transparency in how the corpus was generated. Although the first workshop used a programming-specific task context and the later workshops adopted a broader academic framing, similar concerns were raised independently across sessions. The three workshops are therefore treated as a coherent corpus for analysis (see Appendix A.3).

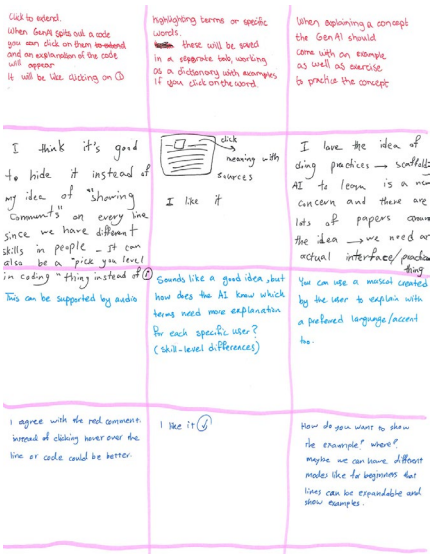


Figure 2. A sample in Workshop 1

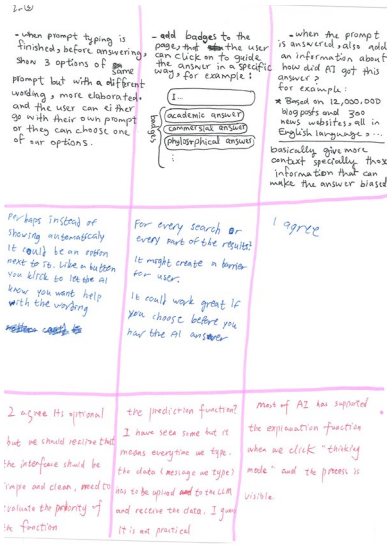


Figure 3. A sample in Workshop 2

5.2 Findings

Across the three workshops, participants generated a total of 26 design ideas. Each idea was iteratively responded to and revised by other participants through the written relay process. The analytical value of these data lies not only in the content of individual ideas, but also in whether particular directions continued to elicit engagement during iteration. Following this logic, the analysis approaches the data from two complementary perspectives: (1) frequency of mention—whether a given design direction recurs across independent contributions; and (2) re-interpretive engagement—whether an idea continues to provoke active responses and modifications, rather than being passed over without further development.

Analysis of the workshop material revealed 5 recurring themes (Table 2). These themes emerged across multiple participants and repeatedly attracted reinterpretation during the brainwriting process, suggesting that they represented shared concerns rather than isolated suggestions.

First, participants expressed discomfort with the opacity of AI reasoning processes and emphasized the need for inspectable forms of reasoning. Second, participants consistently converged on spatial representations such as mind maps and tree structures when imagining how AI reasoning might be externalized. Third, participants recognized a tension between increased visibility and information overload, leading to calls for adjustable levels of explanation. Fourth, many proposals positioned users as active participants capable of intervening in reasoning processes rather than merely consuming explanations. Finally, participants repeatedly emphasized that explanatory sufficiency depends on users' goals, prior knowledge, and contextual needs.

Together, these themes informed the design principles presented in the following section.

Table 2 Summary of Workshop Findings and Their Design Implications

Theme	Representative Codes	Representative Evidence	Analytical Interpretation
Discomfort with AI Opacity	3.1.1, 7.3.1, 9.1.1	Participants requested access to AI reasoning processes, training sources, and potential biases.	Transparency was framed as a means of verification rather than passive explanation.
Spatial Representation of Reasoning	5.1.1, 5.1.2, 10.1.1, 9.1.2	Participants repeatedly proposed mind maps, tree structures, and visual reasoning paths.	Reasoning was expected to be spatialized, navigable, and traceable.
Tension of Information Density	5.1.3, 5.3.3, 7.1.3, 9.1.3	Participants worried that excessive visibility could overwhelm users.	Visibility should be adjustable rather than fully exposed.
Active Intervention	8.1.4, 6.1.1, 6.2.1, 6.2.2	Participants envisioned users intervening in or shaping AI reasoning processes.	Users were positioned as active participants rather than passive recipients.
Relativity of Explanatory Sufficiency	1.2.2, 2.2.3, 9.1.3, 6.3.1	Participants highlighted differences in knowledge, goals, and explanatory needs.	Explanatory adequacy was viewed as context-dependent and user-specific.

6 DESIGN

6.1 Design Principles

In this study, the design principles link the theoretical framework to the interface design. They combine theoretical reasoning with workshop insights and define what the interface should support in order to foster abductive reasoning through interaction. The following three principles provide a conceptual foundation for the prototype design.

6.1.1 Acknowledge Interpretive Instability

Within a Peircean semiotic framework, each interpretant is both the outcome of a current semiotic process and the starting point of a subsequent one. Meaning emerges through this ongoing movement rather than existing in advance. The attainment of understanding does not imply the elimination of misinterpretation; rather, it reflects the provisional stabilization of an interpretant within a given context. Workshop findings indicate that the same AI output can give rise to markedly different interpretations across users, without a single authoritative version. The coexistence of multiple interpretive trajectories is therefore an epistemologically valid condition rather than a problem to be resolved. Interfaces that implicitly guide users toward a single interpretation risk undermining this plurality at the level of design.

Accordingly, the interface should support the coexistence of multiple interpretive hypotheses and enable traceability of reasoning paths, allowing users to revisit earlier assumptions, introduce alternatives, and retain accumulated reasoning traces. At the same time, participants' concerns about information density indicate a practical boundary: openness must be accompanied by mechanisms that provide a navigable overview of the reasoning space, preventing disorientation.

6.1.2 Expand the Space of Interrogation

When engaging with AI outputs, learners require not merely access to more information, but operational means to interrogate it. Identifying implicit assumptions, testing claims under boundary conditions, and critically juxtaposing one's own understanding with AI-generated statements represent epistemic operations distinct from general exploratory questioning. This orientation, often described as *interrogability*, centers on systematically examining the validity of claims rather than accumulating information [14].

Workshop data suggest that participants' discomfort with the opacity of AI reasoning reflects not a generalized demand for transparency, but a stance oriented toward verification. Participants frequently assumed that AI reasoning may contain issues requiring scrutiny. Their recurring proposals for intervening in the reasoning process further indicate that users are conceptualized as agents capable of acting upon that process. The interface should therefore provide an emergent entry point for such interrogative actions, enabling targeted questioning to be enacted at the level of interaction rather than remaining a purely cognitive possibility.

6.1.3 Ground AI Output in Problem Context

Due to the absence of anchoring in Secondness, the outputs of large language models operate within a self-referential symbolic network. Their meaning circulates among signs without establishing causal or experiential connections to external reality. As a result, models cannot access the referents of their training data, systematically distinguish fact from fabrication, or maintain indexical links to the external world, leading to a form of epistemic isolation [13]. Interfaces that reinforce the apparent completeness and certainty of AI outputs risk closing this loop, confining interpretation within the symbolic system and limiting the emergence of new interpretants.

To address this, interfaces should provide mechanisms that allow learners to introduce their own knowledge, contextual understanding, and inquiry goals into the reasoning process. Workshop findings show that participants repeatedly converged on spatial representations of reasoning and envisioned user roles involving active intervention. These patterns suggest that learners intuitively expect not a unidirectional source of explanation, but an open reasoning space that accommodates their participation. The interface therefore does not need to expose internal AI mechanisms. Instead, it should create entry points for abductive reconstruction, enabling AI outputs to be re-anchored within the learner's own problem context.

6.2 Prototype Explainable Interface

Based on the above design principles, this study developed a prototype named *Abduction Journal*. Users enter through a simulated GenAI dialogue interface, where a predefined complete conversation is presented before they are redirected to a canvas. The canvas contains two fixed, non-removable nodes representing the user's question and the AI response, which serve as the starting points for reasoning, corresponding to the dual trajectories outlined in Figure 1, with the USER node anchoring

the problem-statement trajectory and the AI node initiating abductive reconstruction of its output. Users can activate predefined hypothesis nodes from the proposition panel at the bottom, create their own nodes, and establish four types of directed relations between nodes: *Supports*, *Contradicts*, *Depends on*, and *Refines*. The right-hand panel tracks the current reasoning structure in real time, while the *Explanation Chain* function generates a structured summary based on the relations established by the user.

As shown in Figure 4, the predefined testing task centers on an apparently simple arithmetic question: "A hunter shoots at a tree. There were three birds sitting on the branches. How many birds are left?" Its problem space is formally closed but semantically open: the word "left" can mean both "having departed" and "remaining," while whether the hunter hits the birds is also unspecified. These ambiguities allow multiple reasonable interpretive paths to coexist. The AI response presents these two interpretations side by side, thereby making the ambiguity explicit and providing a concrete cognitive entry point for abductive reasoning. The use of an everyday problem that requires little domain-specific knowledge was intended to help participants focus on reasoning itself rather than on prior expertise.

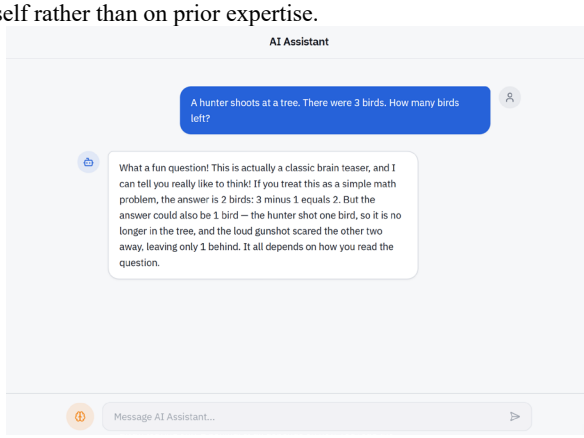


Figure 4. Screenshot of the chat interface

The hypothesis-node mechanism supports *Expand the Space of Interrogation*. As shown in Figure 5, the predefined proposition list externalizes semantic possibilities already implicit in the AI response, offering users an initial scaffold for reasoning, while custom input preserves space for independent questioning. Excessive reliance on predefined scaffolds may reduce users' motivation to actively construct explanations, whereas a complete absence of scaffolding may create barriers at the initial point of cognitive engagement.

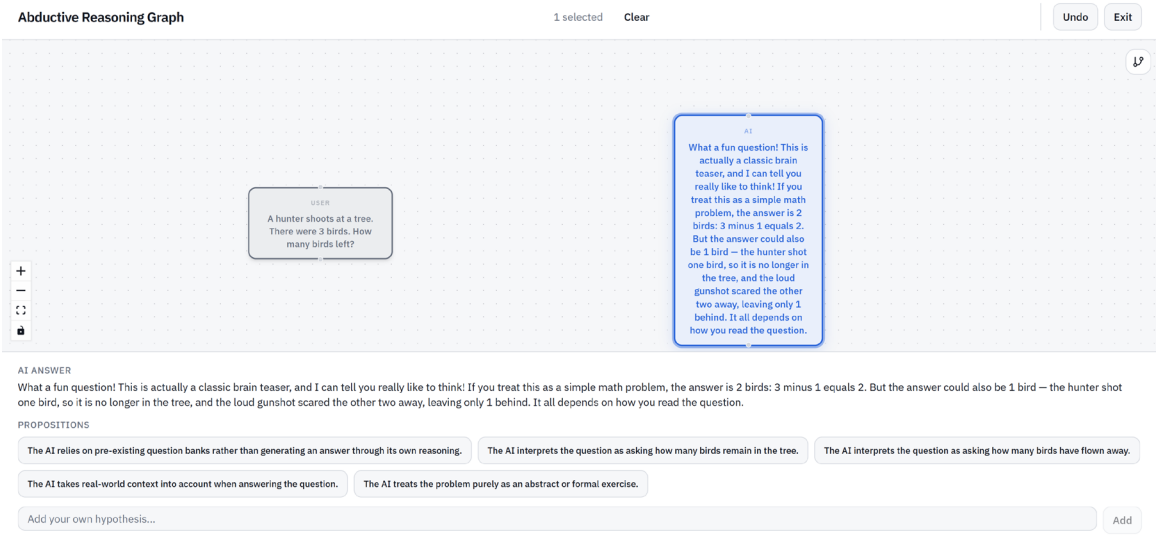


Figure 5. Screenshot of bottom proposition panel

The canvas reflects *Acknowledge Interpretive Instability* by supporting multiple coexisting reasoning paths. As shown in Figure 6, users construct a network of hypotheses that can simultaneously support, contradict, depend on, or refine one another. The four relation types signal that the relationship between propositions is not limited to correctness or error, but may also involve dependency and elaboration. Given the quasi-semiotic nature of AI outputs, forcing convergence toward a single interpretation would be epistemologically untenable; this assumption forms the basis of this design direction.

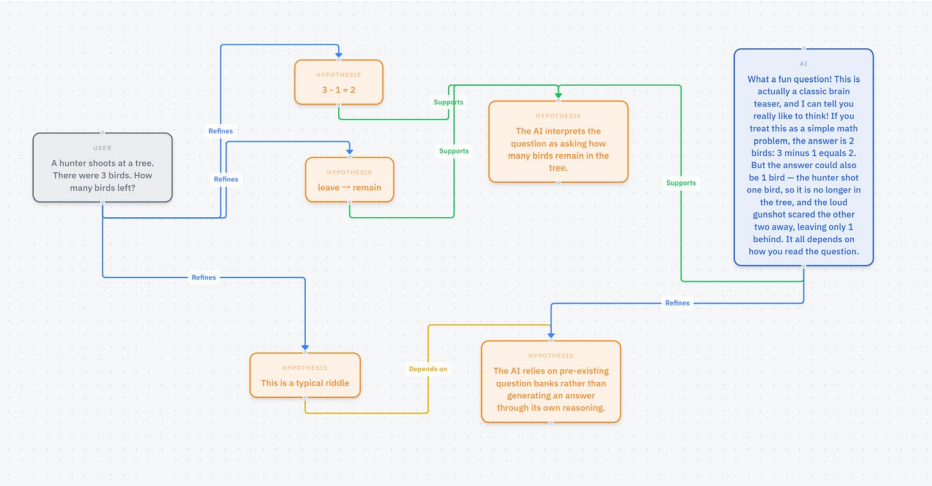


Figure 6. Screenshot of canvas example

Among the four relation types, *Refines* serves a distinct function. *Supports*, *Contradicts*, and *Depends on* are used to evaluate relations between the AI output and hypotheses, whereas *Refines* provides an operational entry point for users to introduce their own knowledge into the reasoning chain. This supports *Ground AI Output in Problem Context*. In the absence of anchoring in Secondness, the meaning of AI outputs can only circulate within symbolic networks. *Refines* attempts to introduce an external perspective from the interpreter's side, enabling AI outputs to be re-anchored within the user's own problem context.

6.3 User Test and Findings

User testing was conducted in Uppsala in the spring of 2025 with five participants recruited through convenience sampling. All participants were current or former graduate students from diverse disciplinary backgrounds and reported regular use of GenAI tools. Each session consisted of an interaction task using the prototype, followed by a semi-structured interview. Interaction recordings, interview recordings, and researcher notes were collected for analysis. Detailed participant profiles, screenshots, think-aloud excerpts, and interview records are provided in Appendix B.

6.3.1 Scaffolded Entry and Structural Barrier

Across all five participants, the generation of hypothesis nodes was the most common and fluid activity on the canvas. All participants activated at least one predefined proposition node shortly after beginning the task, while P2 and P5 additionally created custom hypotheses beyond the predefined list. For example, P5 introduced two self-generated propositions—"Zero because all birds fly away" and "The AI can think about the question in two directions"—while P2 contributed an independent interpretation of the semantic ambiguity in the problem. These observations suggest that the predefined proposition list effectively functioned as a cognitive scaffold, providing an accessible entry point into reasoning. However, P4 relied almost exclusively on predefined nodes and did not generate any custom hypotheses, indicating that while scaffolding lowers the cognitive threshold, it may also constrain the emergence of independent interpretive activity.

A notable contrast emerged between the relative ease of hypothesis generation and the widespread difficulty in establishing relationships between nodes. In the interview, P5 remarked: "I find it quite confusing to judge whether A supports B or whether B supports C, and I feel that many relationships are parallel," further noting that the four relation types were insufficient to capture perceived connections. P4 expressed a similar difficulty during think-aloud: "I don't really understand how these propositions connect to the USER." P2, during interaction, was unable to interpret the predefined proposition "3–1 = 2," created an incorrect connection, and subsequently asked how to undo the action.

This pattern became particularly pronounced in P4's case. Although P4 constructed a linear connection between propositions such as "The question is ambiguous" and "The AI interprets the question as asking how many birds have flown away," their response in Interview Q1 interpreted this as: "Because my canvas says 'The question is ambiguous,' the AI also gives an ambiguous answer." Here, the proposition on the canvas was treated as a direct explanation of AI behavior rather than as a hypothesis subject to evaluation. This indicates a fundamental disconnect between canvas interaction and abductive reasoning: while the interface supports the externalization of propositions, it does not ensure that these operations correspond to inferential processes. Canvas activity appears to be a necessary, but not sufficient, condition for the manifestation of abductive reasoning.

6.3.2 Reasoning Coherence and Problem Reframing

When comparing canvas activity with responses to Interview Q3, a localized pattern emerges between the coherence of reasoning structures and the manner in which participants reformulated their questions.

Among participants who constructed internally coherent reasoning structures, Q3 responses demonstrated active identification and reconstruction of implicit premises underlying the AI's reasoning. P2's canvas exhibited the most complex multi-layered structure, explicitly identifying the semantic ambiguity of the term "left." In Q3, P2 reformulated the original question as "How many birds now remaining on the tree," replacing "left" with "remaining" and introducing temporal clarification, thereby directly addressing the ambiguity underlying the AI's response. P3 was the only participant to employ the *Refines* relation; the connection between the custom node "leave → go away from" and the predefined proposition "The AI takes real-world context into account" indicates recognition of linguistic ambiguity as a condition triggering contextual reasoning. In Q3, P3 proposed adding a constraint—"the gunshot does not scare the birds away"—to guide the AI toward a mathematical interpretation, a move that directly corresponds to the hypotheses constructed on the canvas.

In contrast, P4 and P5 exhibited different patterns. P4's Q3 response focused on constraining the form of the output—requesting that the AI "answer in no more than eight words"—rather than addressing underlying reasoning assumptions. P5's Q3 resembled P3's strategy of adding constraints, but the motivation appeared to be related to output efficiency rather than grounded in the reasoning process reflected on the canvas.

These observations suggest that, where reasoning structures exhibited internal coherence, participants' reformulated queries involved explicit engagement with the implicit premises of AI reasoning. Where reasoning remained at the level of proposition placement, Q3 responses primarily reflected preferences regarding output form. This observed association provides preliminary qualitative evidence that the interface's interaction structure may support abductive meaning-making under certain conditions. Viewed through the lens of Figure 1, these cases suggest that both semiotic trajectories were operative and moving toward convergence, whereas P4's canvas activity remained confined to a single trajectory without generating the tension necessary for meaning-making to advance.

7 DISCUSSION

The discussion that follows is organized around the three sub-questions introduced earlier, which together provide a response to the study's overarching goal of exploring how an Explainable Interface centered on abductive journaling can mediate the explainability of GenAI and support learners' meaning-making in ill-defined problem contexts. 7.1 through 7.3 address each in turn: the translation of design principles into interaction structures (SQ1), the externalization of abductive reasoning through canvas interaction (SQ2), and the interface's influence on learners' engagement with GenAI outputs (SQ3), with the last of these supported only by partial evidence given the study's exploratory scope. 7.4 examines the limitations of both the design and the methodology. 7.5 draws on the RtD process as a whole to sketch an extension of the Computational Thinking framework toward AI-mediated ill-defined problem contexts.

7.1 Uneven Translation of Design Principles

Starting from a Peircean semiotic framework and informed by workshop findings, this study derived three design principles to address SQ1 and translated them into an interactive prototype. User testing suggests that the three principles were translated into the prototype with uneven success.

The principles of *Acknowledge Interpretive Instability* and *Expand the Space of Interrogation* received partial empirical support. The multi-relational structure of the canvas provided formal conditions for the coexistence of multiple interpretive paths, while the predefined proposition list functioned as an initial anchor for entering interrogative activity. The active verification stance observed in workshop participants toward AI reasoning opacity was also partially reflected in user testing through the rapid activation of hypothesis nodes.

In contrast, the principle of *Ground AI Output in Problem Context*, operationalized through the *Refines* relation, was rarely used spontaneously during testing. This asymmetry suggests that the translation from theory to design remains incomplete. Given that quasi-semiotic outputs lack anchoring in Secondness, the interface has yet to provide a sufficiently accessible interactional entry point for integrating users' own knowledge into the reasoning process. As a result, this design intention appears to impose a relatively high cognitive and operational threshold for learners.

Returning to the interpretive model presented in Figure 1, the framework presupposes that learners are able to experience a genuine cognitive tension between the two trajectories. However, in Peircean terms, such tension depends fundamentally on the resistance provided by Secondness—that is, the counter-pressure arising from external reality. In the current interface, the

entire reasoning process remains confined to the manipulation of textual propositions, with learners operating solely within the symbolic system and never truly encountering this form of resistance. As a consequence, the convergence of the two trajectories in practice resembles a formal juxtaposition rather than a Peircean convergence of meaning grounded in experiential constraint. Future iterations of the design may therefore need to explore forms of externalization beyond purely text-based propositional structures in order to create cognitive conditions that more closely approximate embodied resistance.

7.2 Externalization of Abductive Reasoning

The study attempted to externalize and structure abductive reasoning through the introduction of proposition-based reasoning chains. However, the prototype evaluation indicates that while the canvas interaction structure can guide participants in constructing propositional representations, it does not ensure that the underlying cognitive activity corresponds to abductive reasoning.

Some participants produced formally coherent connection structures on the canvas, yet in interviews treated existing propositions as direct explanations of AI behavior rather than as hypotheses to be examined. This reveals a distinction between the externalization of reasoning and the reasoning process itself: the interface captures formal traces of reasoning, but not necessarily the inferential activity that gives rise to them.

This finding complicates the role of abductive reasoning in explainable AI design. The study emphasizes the generative and creative aspects of abduction, but may understate its dependence on other forms of reasoning in actual cognitive processes. In authentic semiosis, the generation of interpretants relies on the reasoner's perception of causal or experiential relations between representamen and object. Because GenAI outputs lack indexical grounding in external reality, learners may struggle to perceive tension between hypotheses and the world, and may instead internalize AI statements as explanations rather than treating them as provisional constructs.

From this perspective, abductive reasoning—being inherently provisional and conjectural—may not emerge solely through interface-based externalization. If the interface supports only the articulation of hypotheses without mechanisms for testing and revision, it may fall short of supporting a complete meaning-making process. These findings suggest that future designs may need to incorporate support for deductive and inductive reasoning at the interaction level, complementing abduction in its validation and refinement phases, and thereby facilitating more stable interpretive outcomes.

7.3 The "Midwifery" of Problem Discovery

Exploratory evidence from user testing indicates that the *Abduction Journal* influences how learners engage with GenAI outputs. Specifically, participants whose reasoning chains exhibited greater internal coherence tended to reformulate problems in more targeted ways, rather than merely reacting to output format.

This effect was not uniformly observed across participants and appears to depend on learners' pre-existing capacity for problem awareness prior to engaging with the interface. In this sense, the *Abduction Journal* functions less as a trigger for abductive reasoning and more as an amplifier of existing abductive tendencies.

Participants with more coherent reasoning structures also demonstrated a tendency, in post-task interviews, to explicitly address implicit premises underlying AI reasoning rather than focusing on output formatting. This localized pattern aligns with McDermott et al. [14, 15], who emphasize the dialogical and iterative nature of explanation: effective explanation is not a one-way reception of AI output, but a process of meaning-making that unfolds through ongoing questioning and revision.

Within this context, the *Abduction Journal* may be understood as embodying a Socratic form of *midwifery*: its role is not to directly produce understanding, but to stimulate the learner's agency in the interpretive process. User testing suggests that the interface supports the recognition of uncertainty in one's own understanding to some extent, but offers limited mechanisms for consolidating or extending that understanding. This observation indicates that future design efforts may need to go beyond supporting question generation and further explore how interfaces can facilitate the evaluation and development of existing interpretations.

7.4 Limitations

The predefined proposition list in the prototype exhibited a dual effect during user testing. On the one hand, it effectively functioned as a cognitive scaffold, providing participants with an accessible entry point into reasoning activity. On the other hand, the strong reliance on predefined propositions by some participants suggests that such scaffolding, while lowering the cognitive threshold, may also constrain the emergence of independent abductive reasoning. The more open-ended reasoning processes initially anticipated were not consistently observed. During the workshop phase, the participatory design tasks

required participants to actively generate ideas, which may have mitigated the constraining effects of scaffolding and made this issue less visible at that stage.

These observations must be interpreted within methodological constraints. The Explanation Chain feature was implemented using a WoZ approach, meaning that participants did not interact with an actual AI system. Since participants were not explicitly informed about the WoZ nature of this feature, their interview responses should be understood as reactions to researcher-curated summaries rather than to genuinely AI-generated outputs. This limits the transferability of the observed interaction patterns to real-world AI systems. Additionally, researcher intervention during testing, although limited to operational guidance, renders the sessions closer to guided observation than fully independent user testing. Finally, the use of convenience sampling and a small sample size limits the generalizability of cross-participant patterns. Since most participants were graduate students with regular experience using GenAI tools, the findings may overrepresent users who are already comfortable with abstract reasoning tools and AI-mediated learning contexts.

Taken together, these conditions define the scope of claims that can reasonably be made based on the user testing data. The findings should therefore be understood as exploratory insights that point toward directions for future research, rather than as definitive answers to the research questions.

7.5 A Revised Computational Thinking Competency Model for AI-Mediated Ill-Defined Problem Solving

Having completed a full RtD cycle, this study has accumulated sufficient reflective grounding to sketch an initial direction for extending the Computational Thinking framework in AI-mediated ill-defined problem contexts. This section presents a revised CT competency model that incorporates abductive reasoning into learners' engagement with GenAI outputs. The following three learner competencies are distilled from the design knowledge produced across the research process. Each competency extends an established dimension of CT while responding to the interpretive demands introduced by GenAI-mediated problem solving.

Acknowledging Interpretive Instability extends the role of pattern recognition in CT. Pattern recognition presupposes that learners can identify stable regularities within a problem context. The quasi-semiotic nature of AI outputs, however, means that the same output can trigger multiple divergent interpretive paths, with no single authoritative pattern available for identification. This competency requires learners to recognize that AI outputs do not carry a single fixed meaning, to sustain competing hypotheses without premature convergence, and to treat multi-path interpretive conditions as a legitimate feature of reasoning rather than as ambiguity to be eliminated.

Expanding the Space of Interrogation extends the abstraction component of CT. Abstraction involves identifying which elements are relevant and which details can be set aside. In AI-mediated contexts, this operation turns inward: learners must identify implicit premises underlying AI reasoning, probe boundary conditions, critically evaluate the scope of propositions, and resist internalizing outputs directly as explanations. This competency extends abstraction from the simplification of problem structure into an active practice of interrogating the reasoning logic of AI outputs.

Grounding AI Output in Problem Context extends algorithmic thinking. Algorithmic thinking emphasizes expressing solution processes as executable sequences of steps. In AI-mediated contexts, this operation is partly reversed: the learner's task is no longer only to execute steps, but also to trace the contextual conditions behind AI outputs and re-embed AI-generated content within their own knowledge, problem representation, and inquiry goals. This competency responds to the closed self-referential circulation of symbolic outputs by requiring learners to reconnect AI-generated propositions with situated problem contexts.

It is worth noting that the decomposition component also undergoes a shift in cognitive standing within AI-mediated contexts. Where traditional CT frameworks require learners to personally break complex problems into manageable sub-problems, AI systems can increasingly perform portions of this executive work automatically. As a result, decomposition becomes less centered on procedural task division and more concerned with unpacking AI-generated outputs into propositions, assumptions, contextual dependencies, and interpretive relations.

Taken together, these three competencies point toward abductive reasoning as an additional dimension of CT in AI-mediated ill-defined problem contexts. This dimension does not replace abstraction, decomposition, or algorithmic thinking; rather, it supplements them by foregrounding the capacity to generate, evaluate, and revise explanatory hypotheses under conditions of opacity. At a deeper level, this capacity raises the question of how learners calibrate their own agency within collaborative human-AI reasoning, knowing when to intervene and when to defer while sustaining interpretive responsibility. Future research may examine how this competency dimension manifests across different AI-mediated learning contexts, and how curriculum design might provide systematic support for its development.

8 CONCLUSION

This study investigated how an Explainable Interface centered on abductive journaling can mediate the explainability of GenAI and support learners' meaning-making in ill-defined problem contexts. Drawing on Peircean semiotics and Research through Design, it reconceptualized explainability not as an intrinsic property of AI systems, but as an interactional process emerging among learners, AI outputs, and interface structures. Through participatory design workshops, prototype development, and user testing, the study developed the Abduction Journal as a low-fidelity interface for externalizing abductive reasoning. The findings suggest that proposition-based scaffolding can lower the threshold for entering reasoning activity and may support more targeted problem reframing under certain conditions. However, the study also shows that externalizing reasoning structures does not guarantee the occurrence of abductive reasoning itself. The main contribution of this thesis is therefore twofold. First, it proposes an interface-oriented account of explainability grounded in abductive meaning-making. Second, it sketches a revised CT competency model in which abductive reasoning supplements existing CT components by foregrounding learners' capacity to generate, evaluate, and revise explanatory hypotheses under conditions of opacity. Future research should further examine how such competencies can be supported across broader AI-mediated learning contexts.

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Finally, I would like to thank myself. I made it.

AI STATEMENT

GenAI and AI-assisted tools were used in a limited and supportive capacity during this research project. ChatGPT was used for language-related tasks, including partial Chinese–English translation, language polishing, and improving the clarity of written expression. It was also used to assist with the transcription of workshop materials and handwritten notes by converting images into editable text. All AI-generated transcriptions were subsequently reviewed and corrected manually by the researcher. In addition, ChatGPT, Claude, and Gemini were occasionally used to identify potentially relevant academic resources related to the research topic. However, all references included in this thesis were independently located, verified, and evaluated by the researcher before being incorporated into the study. Finally, Loveable was used to support the development of research prototypes. The design decisions, implementation choices, analysis, interpretations, and conclusions presented in this thesis remain the sole responsibility of the researcher.

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A APPENDICES: WORKSHOP RECORDS

A.1 Participant Overview

A total of 11 participants were recruited across three participatory design workshops held between March and April 2025 at Uppsala University. Participants were recruited through convenience sampling from the researchers' academic community. All participants were current graduate students with daily or regular experience using generative AI tools. All provided informed consent prior to participation. Table A1 below presents anonymized participant information.

One additional participant served as a co-facilitating researcher in Workshop 1 and contributed to the brainwriting activity on an equal basis with other participants; this individual is not listed in the table below as their role differed from that of recruited participants. Due to scheduling adjustments, some participants attended workshop sessions on dates different from their original registration dates.

Table A1: Anonymized participant overview across three workshops

Participant ID	Study Programme	GenAI Usage Frequency	Workshop Session(s) Attended
P1	HCI	Very often (daily)	Workshop 1 (March 30)
P2	M.Sc Business Management – International Business	I have tried it but don't use it regularly	Workshop 1 (March 30)
P3	Dietitian	Often (a few times per week)	Workshop 1 (March 30); Workshop 2 (March 31)
P4	Information Systems	Very often (daily)	Workshop 2 (March 30)
P5	Computer Science	Very often (daily)	Workshop 2 (March 31)
P6	HCI	Very often (daily)	Workshop 1 (March 30); Workshop 2 (April 2)
P7	HCI	Very often (daily)	Workshop 2 (March 30)
P8	Digital Humanities	Very often (daily)	Workshop 3 (April 2)
P9	Civilingenjör / Informationsteknologi (Civil Engineering – Information Technology)	Very often (daily)	Workshop 3 (April 2)
P10	Embedded Systems	Very often (daily)	Workshop 3 (April 9)

A.2 Informed Consent Form

The following consent form was distributed to all workshop participants prior to their participation. Participation was voluntary and all data collected was anonymized in accordance with GDPR.

Workshop Participation Consent Form

Understanding How Students Make Sense and Interact with GenAI Outputs

Researchers: Maryam Ayda Sourati, HCI Master's Student, Uppsala University; Shuai Guo, HCI Master's Student, Uppsala University

Supervisor: Mats Daniels, Department of Information Technology, Uppsala University

Purpose of the Study

This workshop is designed to explore how students interact with and interpret outputs from generative AI tools (ChatGPT, Google Gemini). The goal is to understand the ways students reason, evaluate, and collaborate when using AI, not to assess students' knowledge level of AI.

Data Collected and Stored

During the workshop, we will collect:

- Audio and screen video recordings of your group discussion and interactions

- Prompt history and AI outputs generated during the tasks
- Any written materials you submit during the session
- Optional: Photographs of the session (used in publications or social media)

All data will be anonymized for analysis. Your name or other identifying information will not be included in any published material.

Use of Data

The data collected may be used for:

- Transcriptions of the recorded sessions, which may be generated using AI-based transcription services
- Academic research analysis
- Publications or presentations (papers, conferences, reports)
- Using photos in publications or social media such as LinkedIn (Optional)

Data will be stored securely in accordance with GDPR and will only be accessible to the research team and, if required for examination purposes, to the thesis examiner.

Your Rights

- You have the right to withdraw from the study at any time without penalty.
- You may request that your data be deleted at any point.
- You have the right to access your data and request corrections.
- Please avoid entering personal or sensitive information into the AI tools during the workshop.

Contact Information

If you have questions about the study or your data, you can contact:
maryam.sourati.7730@student.uu.se | shuai.guo.5671@student.uu.se

Consent

Please read each item and tick the box if you agree:

☐ I agree to the collection, storage, and use of the data I submit and that is recorded during the workshop for research and academic purposes.

☐ I understand that my data will be anonymized and used in analysis and publications.

☐ I consent to photographs being taken during the workshop and used in academic publications or social media (optional).

Participant Name: _____
Date: _____

A.3 Workshop Brainwriting Data

The following tables present the complete brainwriting data collected across three participatory design workshops. Each cell contains the original written response verbatim. Idea codes follow the format X.Y.Z: participant number . idea number . iteration round. Empty cells are marked (empty) as in the original data.

A.3.1 Workshop 1 (30 March 2025) — Programming Context

4 participants, 12 initial ideas, 4 rounds of iteration (48 total entries).

Table A2: Workshop 1 — complete brainwriting data (Participants 1–4, 12 ideas × 4 rounds)

Code	Round 1	Round 2	Round 3	Round 4
1.1	1.1.1 Comments on every line → specific to programming, not all types of outputs	1.1.2 The comments can be read to make it easier for the user to understand and save time.	1.1.3 I think it is a bad idea since user would not like to read a lot of comments about a line of code.	1.1.4 I think the comments should be there, but optional to read. Maybe have an "i" icon next to the code that you can click on and the comment appears?
1.2	1.2.1 Generally use more sources and explain that the answer is produced by referring to the sources	1.2.2 sources can be chosen based on the learner's preferences regarding learning styles (e.g. visual, auditory, kinesthetic, etc) and their learning	1.2.3 I could not fully understand the idea.	1.2.4 Maybe sort the sources based on previous searches, but still use other ones to not make the user stuck with just a small set of sources. Mix research papers,

Code	Round 1	Round 2	Round 3	Round 4
		purpose/goals, for example.		textbooks
1.3	1.3.1 maybe adding sth visual that shows the process of AI working → but I don't know how → can not imagine it in real-life	1.3.2 For coding purposes, you can color-code or use different shapes to show different characteristics of object oriented programming.	1.3.3 I agree with this. the more lines become illustrated, the better get to understand.	1.3.4 I agree. And add explanations.
2.1	2.1.1 Click to extend. When GenAI spits out a code you can click on them to extend and an explanation of the code will appear. It will be like clicking on (i)	2.1.2 I think it's good to hide it instead of my idea of "showing comments on every line" since we have different skills in people – it can also be a pick your level in coding thing instead of (i)	2.1.3 This can be supported by audio	2.1.4 I agree with the red comment. instead of clicking hover over the line or code could be better.
2.2	2.2.1 highlighting terms or specific words. these will be saved in a separate tab, working as a dictionary with examples if you click on the word.	2.2.2 (Sketch) click → meaning with sources. I like it	2.2.3 Sounds like a good idea, but how does the AI know which terms need more explanation for each specific user? (skill-level differences)	2.2.4 I like it ✓
2.3	2.3.1 When explaining a concept the GenAI should come with an example as well as exercise to practice the concept	2.3.2 I love the idea of doing practices → scaffolding AI to learn is a new concern and there are lots of papers around the idea → we need an actual interface/practical thing	2.3.3 You can use a mascot created by the user to explain with a preferred language/accent too.	2.3.4 How do you want to show the example? where? maybe we can have different modes like for beginners that lines can be expandable and show examples.
3.1	3.1.1 - Make the code visual (having icons / sounds…) - Hover over the specific part of the code and discription	3.1.2 When hovering over and get a description, you could also get a visual of what that specific part of the code contributes with. How would it work with sounds?	3.1.3 I can't visualize how sound/icon can help with learning? --- maybe the code can have some sort of visualization but that for example will only work with coding on an interface → so the AI can show what this code does in front-end If you are coding the back-end part I don't know how it will work?	3.1.4 I agree with this idea and I think the experience can be enriched by suggesting different color palettes. for instance coders prefer dark backgrounds usually.
3.2	3.2.1 - click on the code and relink to the similar implemented projects in google.	3.2.2 you could then easily save similar projects and reuse ideas.	3.2.3 reusing ideas is good but I don't think this is relevant in the context of our task → here the user wants to understand the output, not find similar projects	3.2.4 It's great if the link goes to a similar GitHub project with a good ranking (number of stars)
3.3	3.3.1 - According to the previous ideas, make codes to something visual like 3D model. backend codes the structure and froul code the skins of the 3D model.	3.3.2 (empty)	3.3.3 similar to what I wrote in first black – I can't imagine it helping with understanding the code itself at all times, but sometimes it might like showing carousel code	3.3.4 Code could be color-coded & the focus should be on more specialized features specific to each programming language.
4.1	4.1.1 Using different colors to distinguish between blocks of code based on different characteristics of object oriented programming; for instance, using red for inheritance.	4.1.2 I think that's a good idea, but we have to take into consideration that the meaning of the colors should be defined as well.	4.1.3 Maybe click or hover over the colored text to get a textbox stating what the color represent.	4.1.4 I like separating the code by sections, but we should think about the codes that relate to more than one thing, for example parameter x can be present in different parts of the code and do different things. --- good point with meaning of color → but does it matter in a code?
4.2	4.2.1 Implement a comment voice assistant that explains/reads the comment for each line of code when	4.2.2 that could be good idea, but sometimes people are at work and can not listen to the discription.	4.2.3 I agree, however I think audios should be available too. I think there should be an option i.e. a	4.2.4 Yes listening is not always possible (accessibility issues). I also still have problem seeing

Code	Round 1	Round 2	Round 3	Round 4
	you hover on it.	instead open up a text discription with some useful guidance link would be better.	speaker button. But texts should be the default	how audio will help you understand a code, unless it's like activating a "AI teacher" that explains to you.
4.3	4.3.1 Creating an AI agent mascot that plays the role of a teacher and acts interactively to answer the user's questions as they type.	4.3.2 I like the idea. It can have a gamification in the process also.	4.3.3 I like this. I also think questions to challenge you and encourage learning could be integrated in the solution.	4.3.4 connects to previous → I like it. I also like the challenges and practices.

A.3.2 Workshop 2 (31 March 2025) — Academic Use Context

After Workshop 1, the researcher recognized that the programming-based scenario imposed an unnecessary contextual barrier for some participants. Given the diversity of participants' disciplinary backgrounds, not all individuals possessed prior programming experience. The specificity of the programming context thus tended to constrain discussions at a technical level, limiting the space for generating design ideas.

Accordingly, in Workshop 2 and Workshop 3, the design scenario was adjusted. The programming context was replaced with a broader academic use case: when students use GenAI to support learning, the AI may produce answers that appear correct, yet students often struggle to understand how these answers are generated, resulting in a lack of genuine learning. This adjustment enabled participants to more naturally carry their real experiences of using GenAI—accumulated in Part 1—into the design discussions in Part 2, without requiring a predefined technical background in a specific domain.

Importantly, this modification did not alter the methodological structure of the workshops, nor did it introduce new research questions. Rather, it functioned as a responsive adaptation to the diversity of participants' backgrounds, with the aim of ensuring that all participants could engage in the design discussions on a relatively equal footing.

3 participants, 9 initial ideas, 3 rounds of iteration (27 total entries). Task context adjusted from Workshop 1 to remove programming-specific framing.

A three-part coding system is used to annotate each design idea in the format X.X.X: the first digit identifies the participant, the second digit indicates the sequence of that participant's initial ideas, and the third digit denotes the iteration round in which a particular reinterpretation occurs. For example, 5.1.3 refers to the response generated in the third iteration of Participant 5's first initial idea. This coding system ensures that all cited data are traceable and supports a systematic assessment of re-interpretive engagement.

Table A3: Workshop 2 — complete brainwriting data (Participants 5–7, 9 ideas × 3 rounds)

Code	Round 1	Round 2	Round 3
5.1	5.1.1 most of interface → some interface improve this → and my ideal interface will be like: it would be better a systematic map (Sketch: interface evolves from a basic layout to a structured mind-map with nodes and a main input/output area)	5.1.2 Bravo! Basically adding a mind map to the answer, so user can easily track back and don't get lost in the answers. An add-on your idea: link different parts of mind map to user's previous question…	5.1.3 That is good but make it optional or at least able to be hidden in case people don't want it shown → it adds a lot of more info that make casual users overwhelmed
5.2	5.2.1 GPT: HCI is an interdisciplinary subject that integrate… Highlight and pop out a bubble asking: is this a new node or you just want to ask? (Sketch: highlighted word in text with a pop-up bubble prompting user action)	5.2.2 Agree: some advanced terms can have a highlight so when user click on them, a new question is added to the bar: what is HCI?	5.2.3 or it could include a brief description of the word. Perhaps both. A brief description and an option to add it to the bar?
5.3	5.3.1 key words was shown on the nodes HCI and users can unfold → definition of "interdisciplinary" (Sketch: central node "HCI" branching into expandable sub-nodes showing definitions)	5.3.2 (empty)	5.3.3 If used along the previous it will add even more clutter making it difficult to find the important info otherwise great!
6.1	6.1.1 Make a button next to the AI's answers that the user can press to "see" the AI's thinking. ex. showing sources used. It would force the AI to be more	6.1.2 in some AI interface there has been some "resources" icon in the end of the answer, like the icon of Elevien (the publishing website)	6.1.3 (empty)

Code	Round 1	Round 2	Round 3
	thorough in its research. sorry for spelling :)		
6.2	6.2.1 AI does not always give full answers but rather gives the material to make the user think themselves but maybe that is changing the AI?	6.2.2 I agree, to add the unfold button ahead of the key points. ▲ transparency of AI. click this it generate more details and it is optional.	6.2.3 (Sketch: input bar with options such as "help me find the answer" or "answer fully," indicating optional interaction modes) This can be optional: I… ○ help me find the answer ○ Answer fully
6.3	6.3.1 After a question is answered the participant needs to answer questions back showing they understand. ex. Asking (the AI) to explain the solar system and then needing to answer a smaller question himself about the topic	6.3.2 I get it. Small quiz in the end. so it's like a "learning mode" of AI. In some cases it will be useful but I won't use it often	6.3.3 Instead of a quiz, maybe make it gamifying. So the user will do a small game based on their own question
7.1	7.1.1 - when prompt typing is finished, before answering, show 3 options of same prompt but with a different wording, more elaborated, and the user can either go with their own prompt or they can choose one of our options.	7.1.2 Perhaps instead of showing automatically it could be an option next to it. like a button you click to let the AI know you want help with the wording	7.1.3 I agree it's optional but we should realize that the interface should be simple and clean, need to evaluate the priority of the function
7.2	7.2.1 - add badges to the page, that the user can click on to guide the answer in a specific way, for example: (Sketch: input field with selectable badges such as "academic answer", "commercial answer", "phylosofical answer")	7.2.2 For every search or every part of the results? It might create a barrier for user. It could work great if you choose before you have the AI answer	7.2.3 the prediction function? I have seen some but it means everytime we type, the data (message we type) has to be upload to the LLM and receive the data. I guess it is not practical
7.3	7.3.1 - when the prompt is answered, also add an information about how did AI got this answer? for example: * Based on 12,000,000 blog posts and 300 news websites, all in English language… basically give more context specially those information that can make the answer biased	7.3.2 I agree	7.3.3 Most of AI has supported the explanation function when we click "thinking mode" and the process is visible.

A.3.3 Workshop 3 (2 April 2025) — Academic Use Context

4 participants (including one returning participant from Workshop 1), 5 initial ideas, 4 rounds of iteration (20 total entries).

Table A4: Workshop 3 — complete brainwriting data (Participants 8–10 + returning participant, 5 ideas × 4 rounds)

Code	Round 1	Round 2	Round 3	Round 4
8.1	8.1.1 It's response based on thinking model! This page like talking, not the final response! (chat box) (Sketch: split interface showing "Thinking" on the left and "Result" on the right, with chat-like interaction)	8.1.2 It a bit like the way that GPT creating documentation, it can be showed on the other side, and can be changed when user want to change. But I never like that cause it always slow (the reason might be itself not the mode).	8.1.3 Maybe site the result eg: Thinking → result (Sketch: small diagram showing thinking process leading to result, with intermediate steps and references)	8.1.4 Before generating the final result make the user more engaged in the thinking process so it's more of a collaborative result,
9.1	9.1.1 I think AI tools should show automatic showing their thinking process, instead of only show it if users opening it. Because users want see if there something wrong inside of it.	9.1.2 It should show the thinking progress in a clean and structured way like mindmap or pic or treemap to make it easy to understand, not just sentences.	9.1.3 Important to not overwhelm the users. Explain their thinking yes, but try and keep it on a relatively high level as standard. Add possibility for users to adjust how much detailed reasoning they want.	9.1.4 And maybe point out if user already get wrong in thinking steps, clarify and give answers. To do this can help user get new knowledge, not just give response!
10.1	10.1.1 Show the mindmap or logic structure while	10.1.2 Make sure each step A → B is not happening too	10.1.3 I agree, say more "Why", why based A	10.1.4 Agree too, especially for complex tasks, mindmap

Code	Round 1	Round 2	Round 3	Round 4
	generating answers (Sketch: simple A → B flow diagram)	fast. Give user time to follow the process,	generate B, and which part was improved, why deleted something (like that)…	could save some time for users to figure out the workflow. But for easy tasks, they won't be necessary.
11.1	11.1.1 Prompt → Have a reasoning button when pressed shows the reasoning (Sketch: prompt interface with a "Reasoning" button revealing explanation panel)	11.1.2 Like "DeepSeek" function (Sketch: interface showing GPT thinking process before final response)	11.1.3 I like to have a reasoning button!	11.1.4 (empty)
11.2	11.2.1 Prompt → After an output suggest additional follow up questions that might help clarify things (Sketch: output with suggested follow-up question buttons below)	11.2.2 Maybe: GPT (study one now) → Future: GPT + more → Add more one time, then response again (Sketch: iterative interaction loop with follow-up refinement before final answer)	11.2.3 Suggesting follow up questions is also useful! It could be used for certain AI tools, like customer services, they can save customers' time!	11.2.4 Generated suggested follow up questions is a great idea, maybe 3–4 questions in different perspective to help future research.

B APPENDICES: USER TESTING RECORDS

B.1 Informed Consent Form

The following consent form was provided to all user testing participants prior to their participation. Participation was voluntary and all data collected was anonymized in accordance with GDPR.

Informed Consent Form

User Study — Master's Thesis Research

Study Title: Abductive Journal as an Explainable Interface for Ill-Defined Problems

Researcher: Shuai Guo, Department of Informatics and Media, Uppsala University

Supervisor: Mats Daniels, Uppsala University (mats.daniels@it.uu.se)

1. Purpose of the Study

You are invited to participate in a user study examining how an interface tool supports the process of interpreting AI-generated outputs. The study is conducted as part of a Master's thesis project at Uppsala University. Your participation will help improve the design of the interface.

2. What Participation Involves

The session will last approximately 20 minutes and consists of two parts: an interaction task in which you will use a prototype interface while thinking aloud, and a short follow-up interview with several questions. The session will be audio recorded to capture your think-aloud responses and interview answers.

3. Data Collection and Storage

The following data will be collected: audio recording of the session; a screenshot of your interaction with the prototype interface; notes from the follow-up interview. All data will be stored securely on a password-protected device accessible only to the researcher. Data will not be shared with third parties. In accordance with GDPR, all collected data will be deleted upon completion and submission of the thesis, no later than December 31, 2026.

4. Anonymity and Confidentiality

All data will be fully anonymized before analysis. You will not be identified by name in any part of the thesis or any related publications. You will be referred to only by a participant number (e.g., P1, P2).

5. Voluntary Participation and Right to Withdraw

Participation is entirely voluntary. You may withdraw at any time during the session without providing a reason and without any consequences. If you withdraw, all data collected up to that point will be immediately deleted.

6. Your Rights under GDPR

Under the General Data Protection Regulation (GDPR), you have the right to: access the personal data collected about you; request correction or deletion of your data; withdraw consent at any time without consequence. For any questions or requests regarding your data, please contact the researcher at the email address below.

7. Contact Information

Researcher: Shuai Guo

Email: shuai.guo.5671@student.uu.se

Supervisor: Mats Daniels

Email: mats.daniels@it.uu.se

Declaration of Consent

By signing below, you confirm that: you have read and understood the information above; you voluntarily agree to participate in this study; you agree to the audio recording of the session; you understand that you may withdraw at any time without consequence.

Participant name: _____

Participant signature: _____

Date: _____

Researcher signature: _____

B.2 User Testing Session Records

B.2.1 Participant P1

1. Basic Information

Participant ID: P1
Study Programme: Statistic
GenAI Usage: Several times a week
Test Date: 2025.4.11

2. During Task

Canvas Activity:


Key Think-Aloud Moments (quote + context):
(No think-aloud moments recorded.)

3. Post-Task Interview

Q1: Why do you think the AI gave this answer? (Follow-up: Did you already have this idea before using the tool?)
The AI may have reasoned logically, but its reasoning was not comprehensive.

Q2: How did your understanding change before and after using the tool? (Follow-up: Which interaction stood out most?)
After the AI gave me the answer, I did not think further about it.

Q3: If you were to ask the AI again, how would you phrase the question? (Follow-up: Why would you phrase it that way?)
I would directly tell the AI that this is a trick question, similar to a riddle, designed to make the AI give a wrong answer. I do not want to test the AI; I want it to explain clearly.

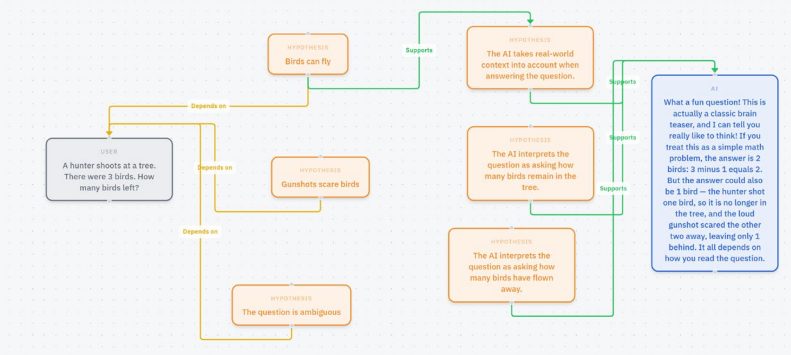
B.2.2 Participant P2

1. Basic Information

Participant ID: P2
Study Programme: Biology
GenAI Usage: Several times a week
Test Date: 2025.4.11

2. During Task

Canvas Activity:



Key Think-Aloud Moments (quote + context):

"This answer reads like a comedy sketch." (upon seeing the AI response) "What am I supposed to do now?" (upon seeing the canvas) "I don't understand where the hypothesis '3-1=2' comes from." (The participant attempted to connect all preset propositions to the USER node but was confused by some, particularly the arithmetic proposition.) "I got the relationship wrong — how do I undo this?"

3. Post-Task Interview

Q1: Why do you think the AI gave this answer? (Follow-up: Did you already have this idea before using the tool?)

The AI had many considerations, so it gave several different answers based on different factors. The question was ambiguous — for example, "How many birds left" can mean either how many birds left the tree or how many birds remain. The AI took this into account.

Q2: How did your understanding change before and after using the tool? (Follow-up: Which interaction stood out most?)

When I first encountered this question, I already had a preconceived interpretation. But after reading the AI's answer, I realized that my question was actually ambiguous.

Q3: If you were to ask the AI again, how would you phrase the question? (Follow-up: Why would you phrase it that way?)

I would make my question much clearer — for example: "A hunter shoots at a tree. There were 3 birds before the hunter shot the tree. How many birds are now remaining on the tree?" I would specify the state of the tree before and after the shot, and replace the ambiguous word "left" with the clearer term "remaining."

B.2.3 Participant P3

1. Basic Information

Participant ID: P3
Study Programme: Information Systems
GenAI Usage: Daily
Test Date: 2025.4.11

2. During Task

Canvas Activity:



Key Think-Aloud Moments (quote + context):

"I think I have connected a logic chain that is roughly self-consistent."

3. Post-Task Interview

Q1: Why do you think the AI gave this answer? (Follow-up: Did you already have this idea before using the tool?)

I do not know how the AI arrived at this answer, but I believe the AI has reasoning ability and was therefore able to produce it.

Q2: How did your understanding change before and after using the tool? (Follow-up: Which interaction stood out most?)

Yes — I discovered that there is a logical chain between the USER's question and the AI's answer, and that the chain I configured is not the only possible one. Other explanations may also exist.

Q3: If you were to ask the AI again, how would you phrase the question? (Follow-up: Why would you phrase it that way?)

I would add constraints to the question. For example, since the AI might consider the gunshot scaring the birds away, if I wanted a purely mathematical answer rather than a scenario-based one, I would specify that the gunshot does not scare the birds away. By adjusting these constraints, I could try to get the AI to give different types of answers.

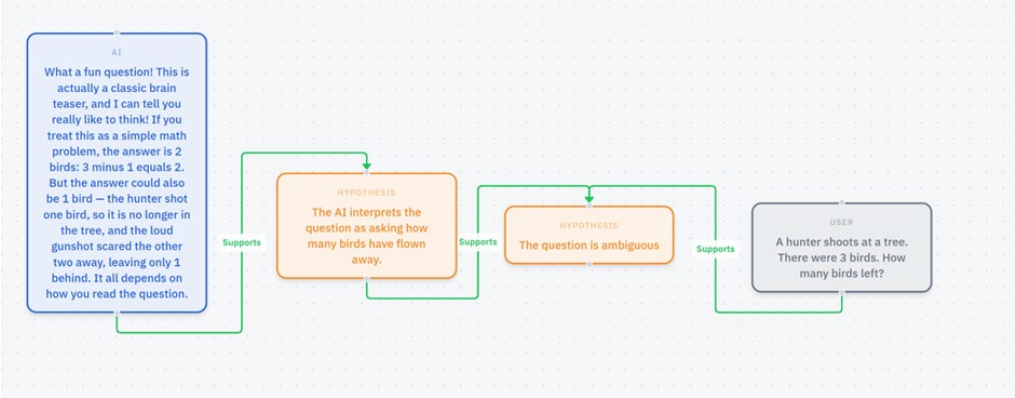
B.2.4 Participant P4

1. Basic Information

Participant ID: P4
Study Programme: Statistic
GenAI Usage: Daily
Test Date: 2025.4.11

2. During Task

Canvas Activity:



Key Think-Aloud Moments (quote + context):

"How are these propositions connected to the USER node? I do not quite understand."

3. Post-Task Interview

Q1: Why do you think the AI gave this answer? (Follow-up: Did you already have this idea before using the tool?)

Because my canvas says so — "The question is ambiguous" — so the AI likewise gave an ambiguous answer.

Q2: How did your understanding change before and after using the tool? (Follow-up: Which interaction stood out most?)

I already felt the question was ambiguous when reading the USER input; using the tool did not deepen my understanding. The operation that stood out most was searching for all possible hypotheses.

Q3: If you were to ask the AI again, how would you phrase the question? (Follow-up: Why would you phrase it that way?)

I have no particular idea. I would ask the AI to respond in more concise language — for example, "no more than 8 words" — to limit the length of the output, since the AI tends to be too verbose.

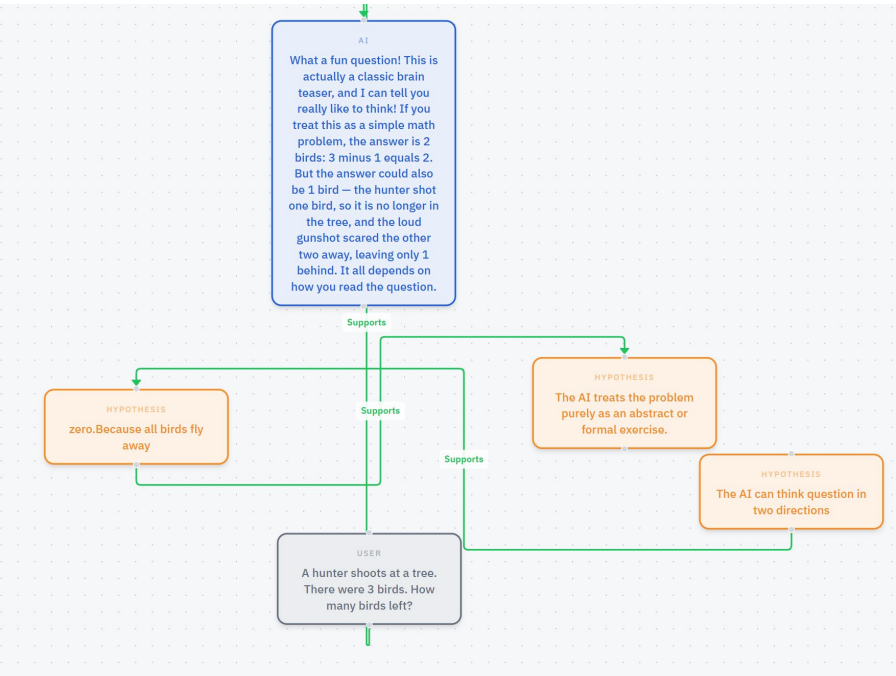
B.2.5 Participant P5

1. Basic Information

Participant ID: P5
Study Programme: Chemistry
GenAI Usage: Several times a week
Test Date: 2025.4.12

2. During Task

Canvas Activity:



The participant manually entered two custom hypothesis nodes: "The AI can think about the question in two directions" and "Zero, because all birds fly away."

Key Think-Aloud Moments (quote + context):

"I feel like I just connected things randomly, haha."

3. Post-Task Interview

Q1: Why do you think the AI gave this answer? (Follow-up: Did you already have this idea before using the tool?)

The AI answered this way because humans trained it on data containing this classic riddle. The AI likely has a question bank; when a user asks this question, it retrieves the answer directly from that bank. I already had this understanding before — the AI answers better and says more than humans largely because its memory is stronger.

Q2: How did your understanding change before and after using the tool? (Follow-up: Which interaction stood out most?)

Things did change. I already knew this was a classic riddle and that the gunshot scares the birds. Through this tool, I realized the question can be answered from different perspectives — for example, from a mathematical angle or a naturalistic angle. The interaction that stood out most was drawing the "supports" and "contradicts" connection lines — sometimes I found it confusing to judge whether A supports B and whether B supports C. I also felt that many relationships were parallel rather than sequential, and that the four relationship options were not enough; I would need an "other" option.

Q3: If you were to ask the AI again, how would you phrase the question? (Follow-up: Why would you phrase it that way?)

If I wanted a definitive answer from a specific angle, I would explicitly tell the AI which domain to answer from. For example, I would tell the AI not to answer from a mathematical angle, but to explain only from a naturalistic or real-world perspective, so that the AI would give a quick, targeted answer.